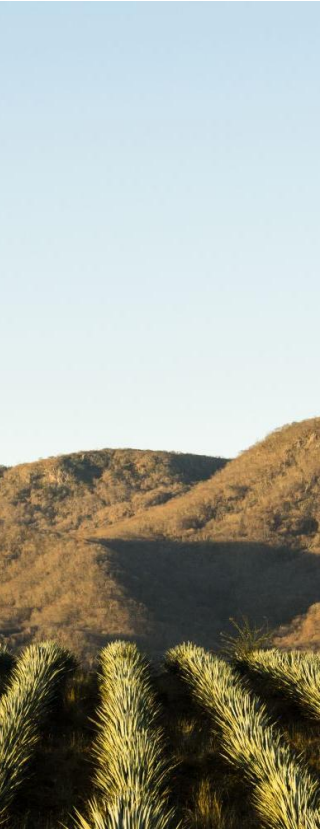


Grassland Fire Impacts on Soil Properties and Vegetation Regrowth

Emma Mott

Introduction

Grassland fires can drive soil properties such as temperature, moisture, and nutrients, which then influence the growth of vegetation.

This is especially relevant right now as the severity and frequency of fires is increasing due to human-induced climate change.

My study gap: The compounding effects of fire and warming on soil properties

Research Questions

CRQ: How do grassland fires and post-fire warming influence soil properties and vegetation recovery over time in a grassland site in Tilden Regional Park?

1) How does soil moisture, temperature, and electrical conductivity differ between burned, nonburned, and burned + heated plots over time?

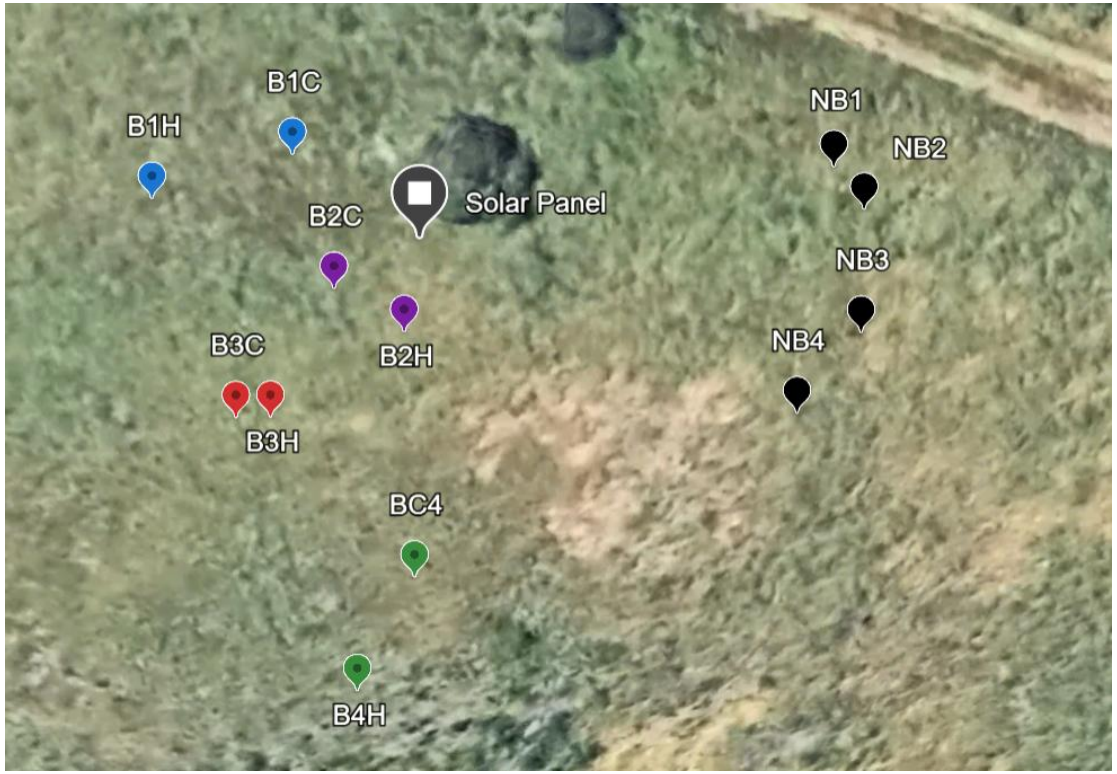
2) How does vegetation regrowth differ across these plots?

3) Are differences in vegetation recovery associated with changes in soil properties?

Project Approach

- Soil Data Collection
- Analysis: Linear Mixed Effects Modelling?
- NDVI Data Collection
- Analysis: ANOVA- NDVI vs Treatment
- Analysis: Multiple Linear Regression- NDVI vs Soil Properties

Study Site: Tilden Regional Park

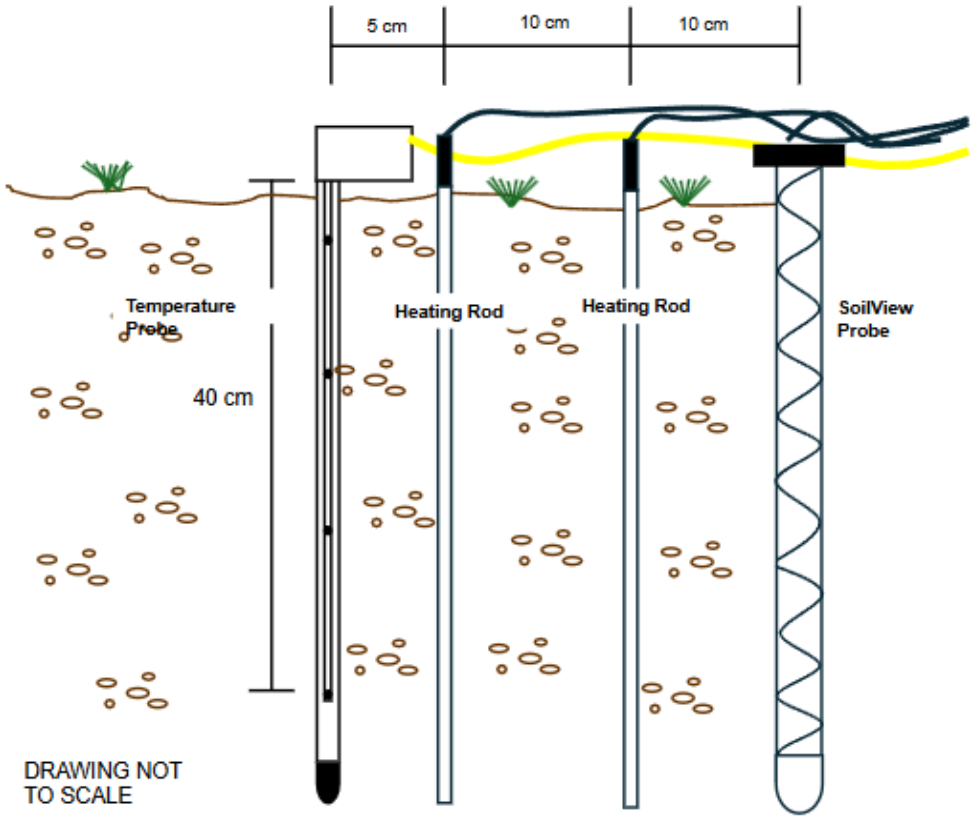


Plot set-up

Area of burn, with study site circled

-Small grassland fire in August 2024
- Experiment installed by LBNL team in October 2024

Soil Plots- Sensors



Schematic of a heated plot

Campbell Scientific SoilVue 10 Sensor – Reads moisture and temperature at multiple depths

Heating rod

Pilot Study

Fall 2025

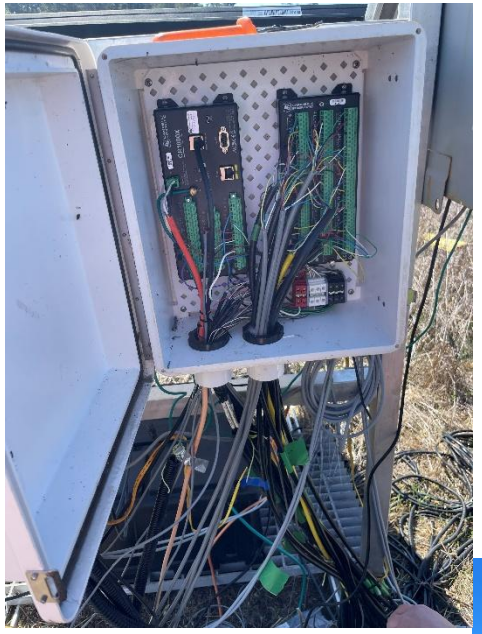
Objectives

Visit field site for observations

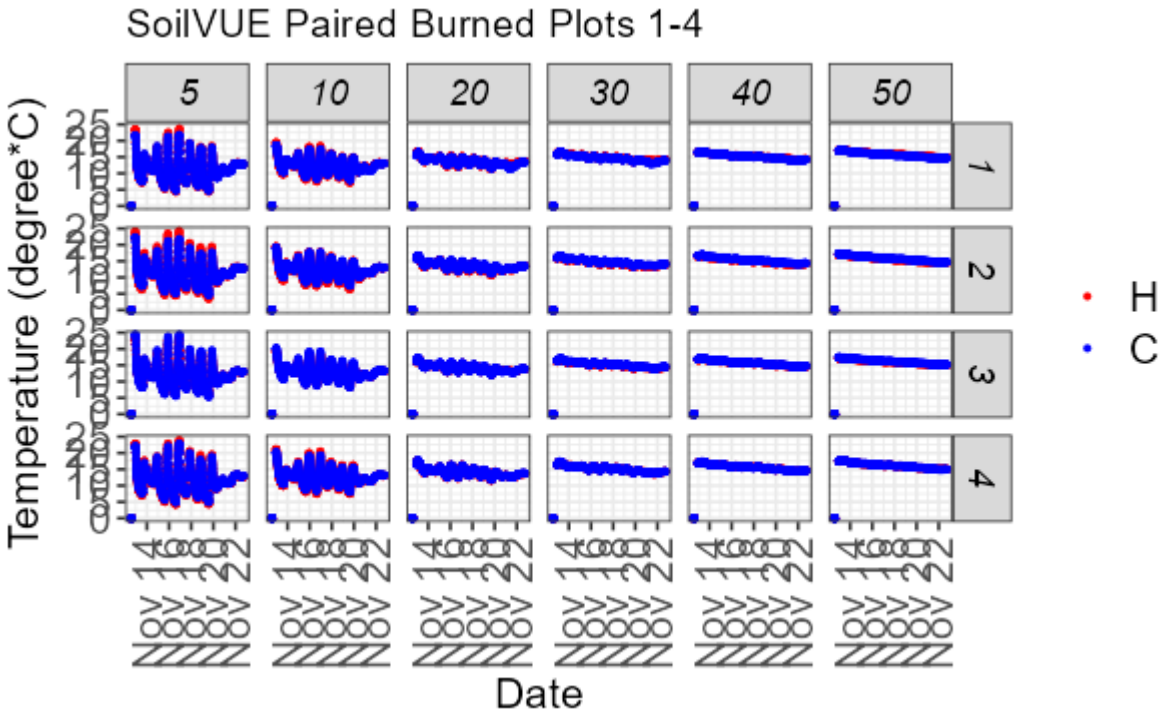
Perform Linear Mixed Effects model on soil data

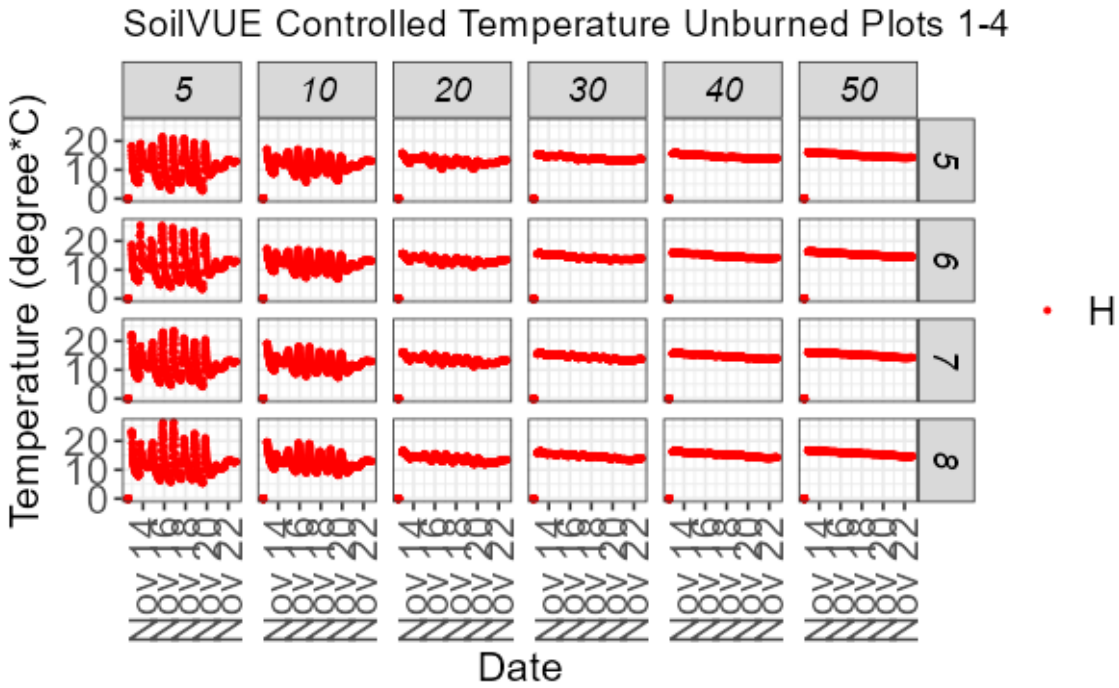
Calculate NDVI values for site

Field Site Visit



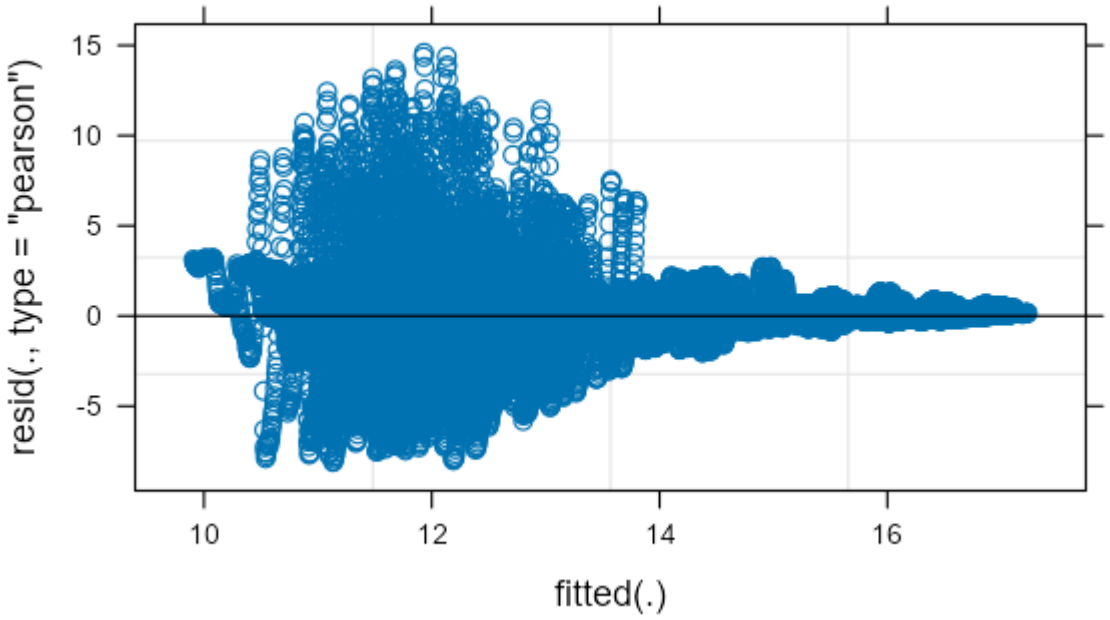
Results- Soil Temperature Data

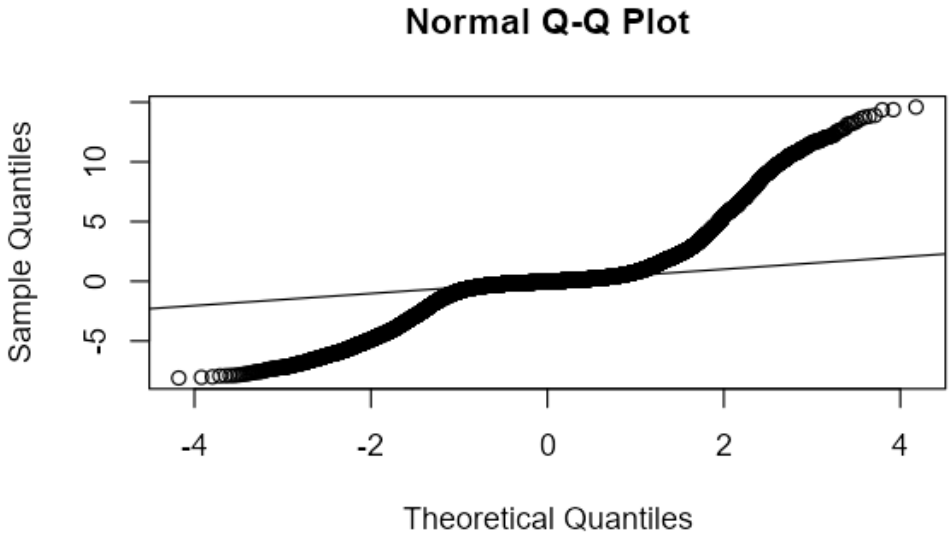




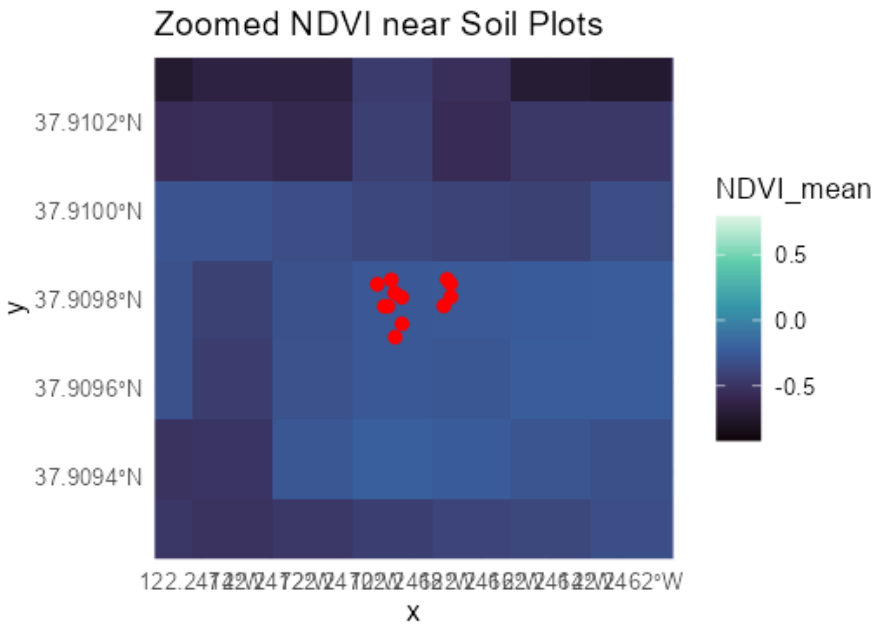
Trouble in Paradise

```
model_temp <- lmer(temp_num ~ trt * depth +
time_z + (1 | PP), data = df)
```

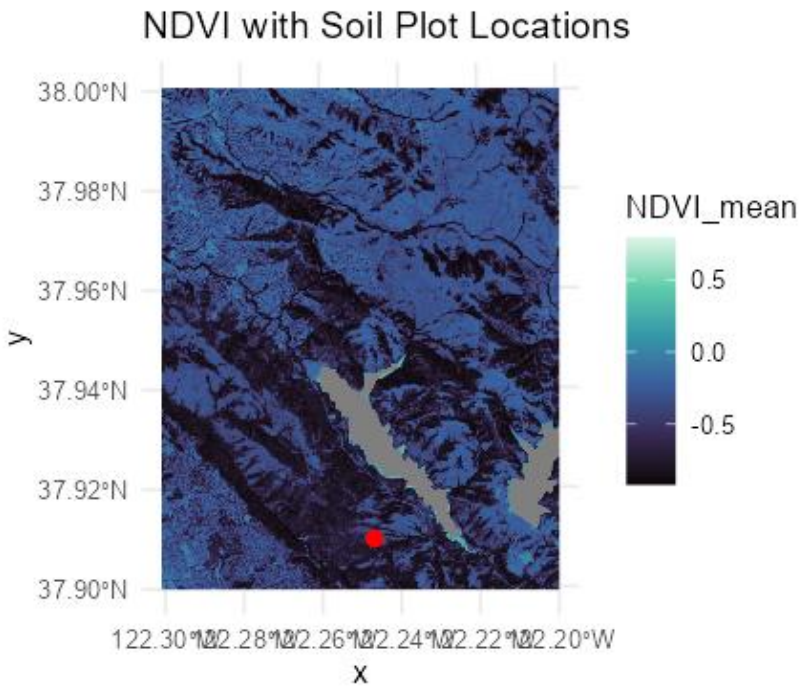




Results- NDVI Calculations



- Too close?



- Too far?

Next Steps

Because the assumption checks showed violation of normality and equal variance, I will have to pivot from the LMM data method

I will attempt one-way ANOVA and multiple linear regression next

For NDVI analysis, I will find better resolution images for my bounding box

Acknowledgements

- Thank you to the team at Lawrence Berkeley National Lab, including my mentors Jackie Pena and Hunter Seubert.
- Thank you to Professor Carl Boettiger for help with the NDVI calculations.

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